INCH-POUND

A-A-55270
April 17, 1995

COMMERCIAL ITEM DESCRIPTION

UNDERWEAR SET, LIGHTWEIGHT, COLD WEATHER

The General Services Administration has authorized the use of this commercial item description for all Federal agencies.

1. SCOPE

1.1 This commercial item description covers the requirements for a lightweight cold weather underwear set. The underwear set is intended for wear by male and female personnel of the Department of Defense.

2. SALIENT CHARACTERISTICS

2.1 Design and construction. The underwear set shall be a two-garment set that shall provide its user with a lightweight layer of insulation in a cold weather environment. The components shall be a lightweight undershirt and lightweight drawers. Both garments shall be composed of 100 percent polyester material with the following properties:

a. Weight - Fabric shall weigh no less than 4.2 oz./yd.^{2} and no more than 5.5 oz./yd.^{2} when tested in accordance with ASTM-D3776, Option C.

b. Fabric count - Rib knit 1 x 1. Minimum fabric count (wale/course) shall be 24/24 when tested in accordance with ASTM-D3887.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FQSC, 2800 South 20th Street, Philadelphia, PA 19145-5099.

AMSC N/A

FSC 8415

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c. Air permeability - Minimum air permeability shall be 450 $ft.^3/min./ft.^2$ when tested in accordance with ASTM-D737.

d. Burst strength - Minimum burst strength shall be 70 lbs. when tested in accordance with ASTM-D3787.

e. pH - Fabric pH must be higher than 6 and lower than 8.5 when tested in accordance with FED-STD-191.

f. Dimensional stability - Fabric shall not shrink or elongate more than 10 percent in the wale or course direction, after undergoing 15 laundry cycles, when tested in accordance with AATCC-135 (1) VAi.

g. Water absorption - Dynamic water absorption when tested per AATCC-70. Fabric to have a finish that imparts permanent soil release, hydrophilic properties.

The underwear set shall be resistant to mildew and shall not retain odors. The underwear components shall have a soft hand that shall prevent chafing and shall retain their shape and softness after repeated laundry cycles. The color of the lightweight cold weather underwear set shall be Brown 436.

h. Matching. The color of the finished cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K.

i. Colorfastness. Colorfastness tests and requirements are indicated in the following table:

Characteristic	Requirement	Test method
Colorfastness to:		
Crocking	Equal to standard sample or not lower than Grade 4.	AATCC TM-8
Laundering (3 cycles)	Equal to standard sample or not lower than Grade 4 for color change. Do not rate staining.	AATCC TM-61, 3A 10 steel balls, 4 gram sample
Perspiration, acid and alkaline	Equal to standard sample or not lower than Class 4 for color change or staining.	AATCC TM-15

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2.1.1 Drawers. The drawers shall extend from the user's waist to a level below his/her ankles and end with elastic ankle cuffs. The ankle cuffs shall consist of double thickness of the basic material. The ankle cuffs shall have sufficient elasticity to prevent constriction of the blood flow to the feet. The waist band shall be made of an elastic hydrophobic, synthetic material. The drawers shall have a standard fly hole in the front. The drawers shall show no obvious signs of deterioration such as broken threads, open seams, frayed hems, holes in fabric due to ruptured fibers, or frayed overedged edges after 15 wash cycles.

2.1.2 Undershirt. The undershirt shall have long sleeves with elastic wrist cuffs. The wrist cuffs shall consist of double thickness of the basic fabric. The wrist cuffs shall have sufficient elasticity to prevent constriction of blood flow to the hands. The undershirt shall have a tubular collar (turtle neck) that extends a minimum of 4 inches above the shoulder level. The undershirt shall have a zipper that begins in the center of the collar top and extends to a level near the bottom of the user's chest. The zipper shall be designed for easy operation with a gloved hand, and shall provide the user with ventilation during strenuous activity and shall be a good match to the basic shade. The undershirt shall show no obvious signs of deterioration such as broken threads, open seams, frayed hems, holes in the fabric due to ruptured fibers, or frayed overedge edges after 15 wash cycles.

2.2 Thread. The thread shall be stretch poly-textured size Tex (T) 24, 45 or 50 or any combination thereof, and size 35 loop/bobbin of A-A-52095.

2.3 Sizes. The underwear set shall come in five standard commercial sizes: extra small, small, medium, large, and extra large. The sizes may be abbreviated as XSM, SM, MED, LG, and XLG.

2.4 Wash cycles. Wash cycle referred to in this document shall be conducted with a commercially available home washer and drier. Wash temperature shall be 140°F; rinse temperature shall be 105°F using normal cycle. Drying temperature shall be 140°F to 160°F using normal tumble dry cycle in accordance with AATCC-135 (1) VAi.

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2.5 Labels. A commercial-type label is acceptable provided that an additional printed cloth label containing the contractor's name, contract number, size and National Stock Number (NSN) is included. The commercial label shall include fiber and laundering information. For drawers, the commercial label shall be centered over the inside back closing seam. For undershirt, the commercial label shall be centered on the inside center back of the undershirt at the turtle neck seam area.

2.5.1 Label/tag. Each item shall be individually bar-coded with a paper tag for personal clothing items. The paper tag shall be standard bleached sulfate having a basis weight of 100 pounds. The paper used for the tags shall have a smooth finish to accept thermal transfer and direct printing. The tags shall have a hole and shall be attached to each item by a fastener, clearly legible and readable by a scanner. The bar coding element shall be a 13 digit national stock number (NSN). The bar code type shall be a medium to high code density and shall be located so that it is completely visible on the item when it is folded and/or packaged as specified and so that it causes no damage to the item.

2.6 Patterns. The contractor shall provide commercial patterns. The patterns shall be approved for use by the Government and shall conform to the cited design and finished measurements.

2.7 Figures. The Figures 1 and 2 are furnished for information purposes only. When inconsistencies exist between the written commercial item description and the figures, the written commercial item description shall govern.

2.8 Sizes and measurements. The finished lightweight cold weather underwear set shall conform to the measurements cited in Tables I and II.

Table I. - Drawers

Size	Overall Length (A)	1/2 Waist (B)	Fly Opening (C)	Cuff Length (D)	Cuff Width (E)
X Small	42-1/2	13	4-1/2	2-1/2	4
Small	43-1/2	14	4-1/2	2-1/2	4
Medium	44-1/2	15	4-1/2	2-1/2	4
Large	45-1/2	16	4-1/2	2-1/2	4
X Large	47	18-1/2	4-1/2	2-1/2	4
Tolerance	± 1	± 1	± 1/4	± 1/4	± 1/4

NOTE: Refer to Figure 1 for A, B, C, D, and E measurements.

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Table II. - Undershirts

Size	1/2 Chest (A)	Length (B)	Sleeve Length (C)	Shoulder Length (D)	Cuff Length (E)	Cuff Width (F)	Collar Width (G)
X Small	15-1/2	28	25-1/2	4-1/2	2-1/2	3-1/2	4
Small	17-1/2	29	26	5	2-1/2	3-1/2	4
Medium	19-1/2	30	26-1/2	5-1/2	2-1/2	3-1/2	4
Large	21-1/2	31	27	6	2-1/2	3-1/2	4
X Large	23-1/2	33	28	6-3/4	2-1/2	3-1/2	4
Tolerance	± 1	± 1	± 1	± 1/4	± 1/4	± 1/4	4

NOTE: Refer to Figure 2 for A, B, C, D, E, F, and G measurements.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.2 Market acceptability. The item offered, or a generic equivalent, must have been sold to the commercial market or to the Government for at least 2 years.

4.3 Visual examination. The finished lightweight cold weather underwear set shall be examined for defects listed in below:

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Defects. Any hole, cut, or tear, needle chew, mend, or thin area; stubby yarn, more than twice the size of the size of the normal yarn; any loose knitting, causing sleaziness; pulled or snagged stitches; any spot, stain, or foreign matter; objectionable odor; tread ends not trimmed; components not cut or fabricated in accordance with this commercial item description; missing, misplaced, incorrect or illegible labels; any seam twisted, puckered or pulled loose or tight tension resulting in an unsatisfactory seam; part of garment caught in any unrelated operation or stitching; any open seams, any raw edge protruded more than 1/4 inch from stitching line or protruding more than 1/8 inch along 1-inch or more of seam; any runoff of more than a 1/4 inch; any variation not otherwise classified herein but at variance with the commercial item description; bar code omitted or not readable by scanner; human-readable interpretation (HRI) omitted or illegible; bar code not visible on folded, packaged item; bar code causes damage to the item; any items not packaged in accordance with the contract or purchase order.

4.4 Quality performance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with ANSI/ASQC Z1.4.

4.5 Classification of inspection. The following types of inspections shall be conducted on this item:

a. Quality conformance inspection

4.6 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with ASTM-D3951.

5.2 Palletization. When specified, shipments of lightweight cold weather underwear sets shall be palletized on a 4-way entry pallet in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with MIL-STD-147.

6. MARKING. Marking on shipping containers shall be in accordance with MIL-STD-129.

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7. NOTES

7.1 Source of Government documents. Copies of military and Federal documents are available from:

Standardization Documents Order Desk
Bldg. 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

7.2 Source of non-Government documents

ASTM Test Methods

(Applications for copies should be addressed to American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AATCC Test Method

(Applications for copies should be addressed to American Association of Textile Chemists and Colorists (AATCC), P.O. Box 12215, Triangle Park, NC 27709-2215.)

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Applications for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3308.)

MILITARY INTERESTS:

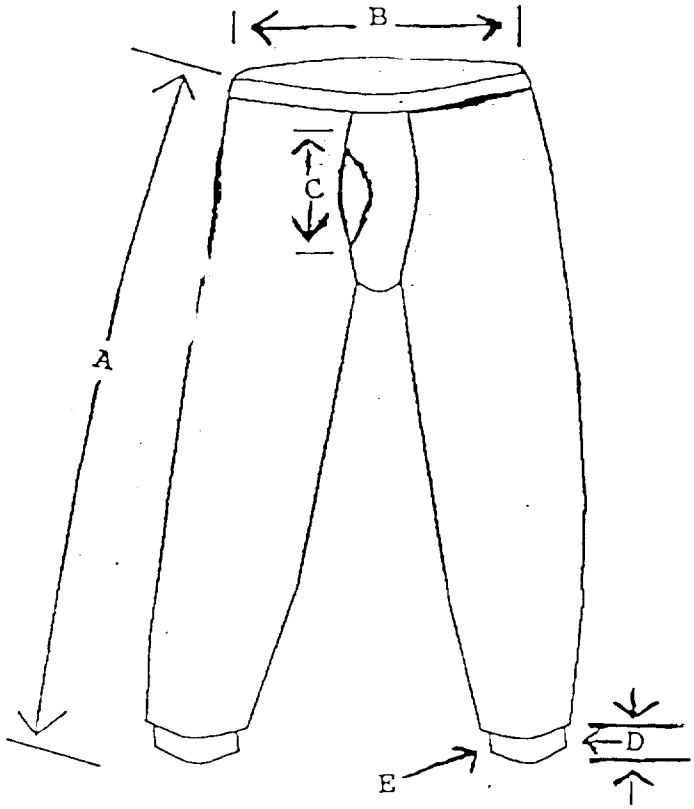
Custodian:
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CIVIL AGENCY COORDINATING ACTIVITY:
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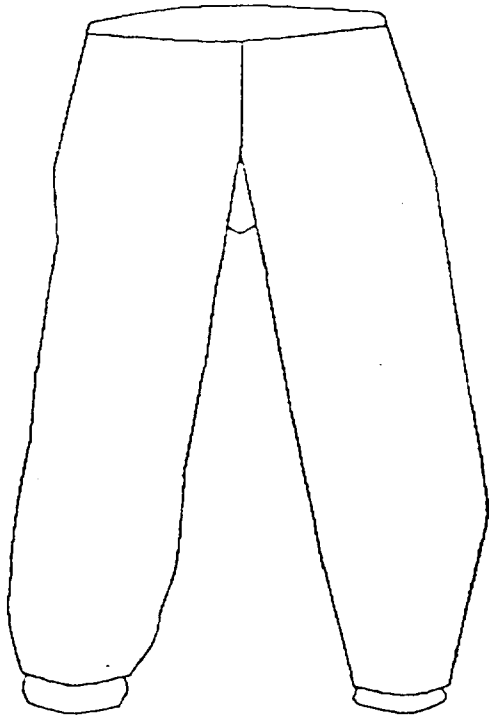
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FRONT
VIEW



BACK
VIEW

FIGURE 1. DRAWERS: UNDERWEAR SET, LIGHTWEIGHT, COLD WEATHER

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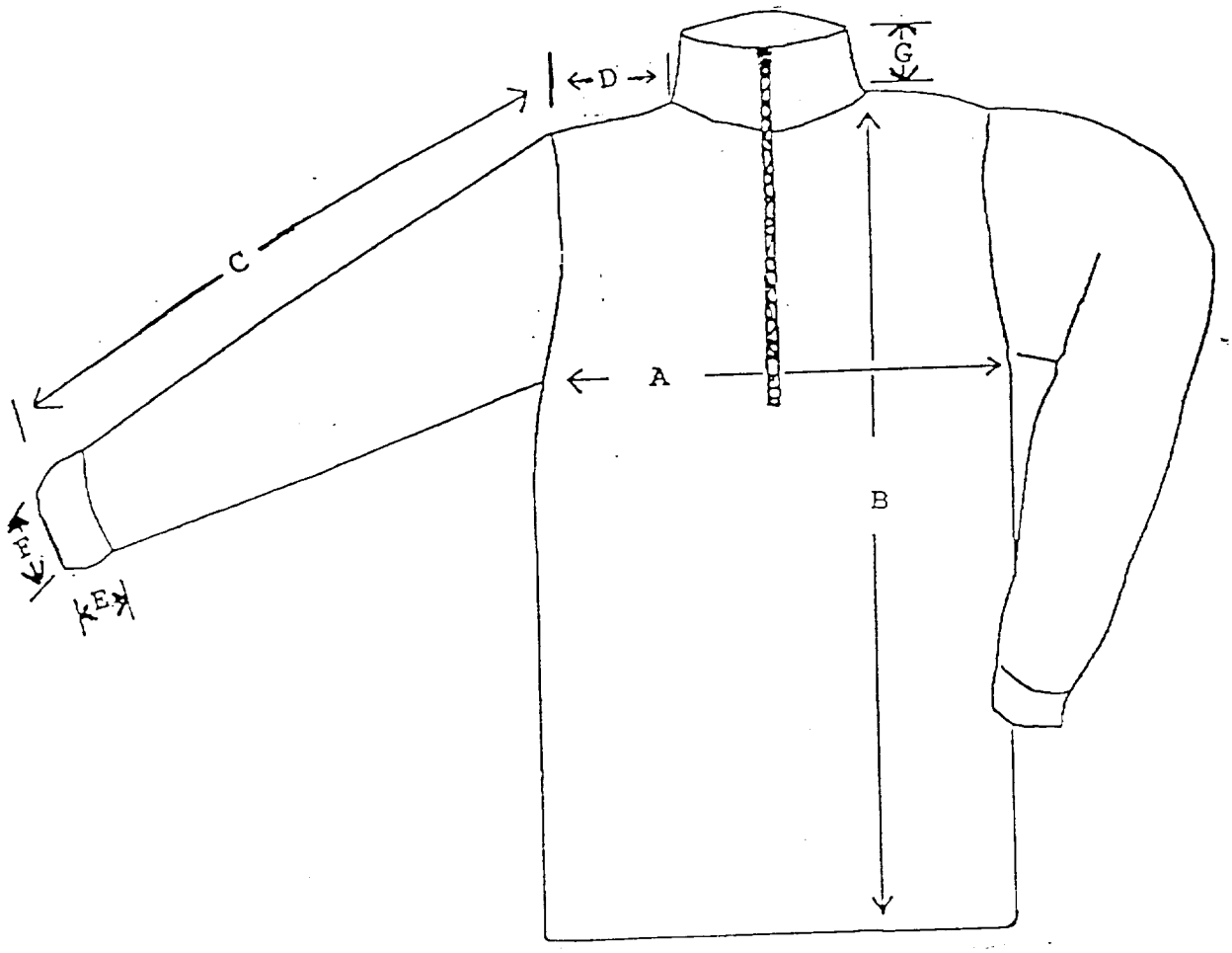


FIGURE 2. UNDERSHIRT: UNDERWEAR SET, LIGHTWEIGHT, COLD WEATHER

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